

Вывод всех тригонометрических формул

Формулы угловения

$$\sin(x + \pi) = -\sin x$$

$$\cos(x + \pi) = -\cos x$$

$$\sin(x + \frac{\pi}{2}) = \cos x$$

$$\cos(x + \frac{\pi}{2}) = -\sin x$$



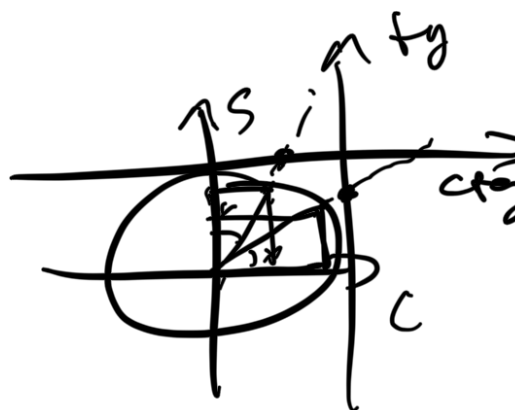
Формулы дополнения

$$\sin(\frac{\pi}{2} - x) = \cos x$$

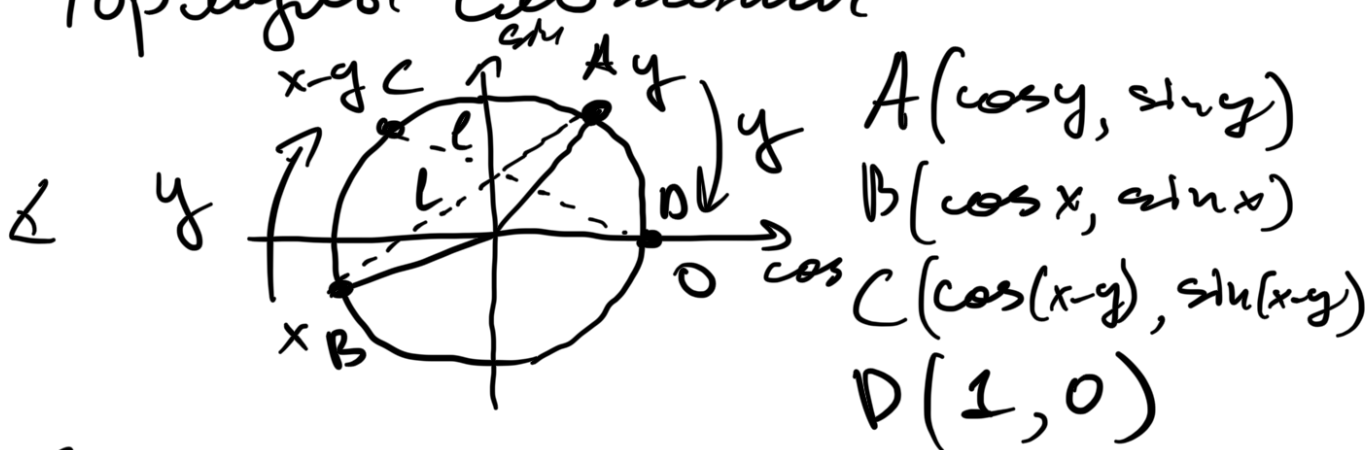
$$\cos(\frac{\pi}{2} - x) = \sin x$$

$$\operatorname{tg}(\frac{\pi}{2} - x) = \operatorname{ctg} x$$

$$\operatorname{ctg}(\frac{\pi}{2} - x) = \operatorname{tg} x$$



Формулы сложения



$$I) l^2 = |\overline{BA}|^2 = |\overline{CD}|^2 \Rightarrow$$

$$l^2 = (\cos x - \cos y)^2 + (\sin x - \sin y)^2 =$$

$$= \cos^2 x + \cos^2 y - 2\cos x \cos y +$$

$$+ \sin^2 x + \sin^2 y - 2\sin x \sin y =$$

$$= 2 - 2(\cos x \cos y + \sin x \sin y)$$

$$\begin{aligned} \text{ii) } l^2 &= (\cos(x-y) - 1)^2 + (\sin(x-y) - 0)^2 = \\ &= \cos^2(x-y) + 1^2 - 2 \cdot 1 \cdot \cos(x-y) + \\ &+ \sin^2(x-y) = 2 - 2 \cos(x-y) \Rightarrow \end{aligned}$$

$$2 - 2(\cos x \cos y + \sin x \sin y) =$$

$$= 2 - 2 \cos(x-y) \Rightarrow$$

$$\cos(x-y) = \sin x \sin y + \cos x \cos y \quad \textcircled{v}$$

$$\cos(x+y) = \cos(x - (-y)) =$$

$$= \sin x \sin(-y) + \cos x \cos(-y) \Rightarrow$$

$\swarrow$ $\searrow$
 $-\sin y$ $\cos(y)$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y \quad \textcircled{v}$$

$$\sin(x+y) = \cos\left(\frac{\pi}{2} - (x+y)\right) =$$

$$= \cos\left(\left[\frac{\pi}{2} - x\right] - y\right) =$$

$$= \cos\left(\frac{\pi}{2} - x\right) \cos y + \sin\left(\frac{\pi}{2} - x\right) \sin y =$$

$\swarrow$ $\swarrow$
 $\sin x$ $\cos x$

$$= \sin x \cos y + \sin y \cos x$$

$$\sin(x+y) = \sin(x-(-y)) =$$

$$= \sin x \cos(\cancel{-y}) + \sin(\cancel{-y}) \cos x \Rightarrow$$

$\cos y \qquad -\sin y$

$$\sin(x \pm y) = \sin x \cos y \pm \sin y \cos x$$

Формулы двойного угла

$$\sin(2x) = \sin(x+x) = 2 \sin x \cos x$$

$$\cos(2x) = \cos(x+x) = \cos^2 x - \sin^2 x$$

$$\begin{aligned} \text{tg}(2x) &= \frac{\sin 2x}{\cos 2x} = \frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x} = \\ &= \frac{2 \sin x \cos x}{\frac{\cos^2 x}{\cos^2 x} - \frac{\sin^2 x}{\cos^2 x}} = \frac{2 \sin x \cos x}{1 - \frac{\sin^2 x}{\cos^2 x}} = \frac{2 \sin x \cos x}{1 - \tan^2 x} \end{aligned}$$

Формулы понижения степени

$$\cos(2x) = \cos^2 x - \sin^2 x \Leftrightarrow$$

$$= \cos^2 x - 1 + \cos^2 x = 2 \cos^2 x - 1 \Leftrightarrow$$

$$= 1 - \sin^2 x - \sin^2 x = 1 - 2 \sin^2 x \Rightarrow$$

$$\cos^2 x = \frac{\cos(2x) + 1}{2}$$

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

Формулы универсальных замен

$$t = \tan\left(\frac{x}{2}\right)$$

$$\rightarrow \tan(2x) = \frac{2 \tan x}{1 - \tan^2 x} \Rightarrow$$

$$\tan(x) = \frac{2 \tan\left(\frac{x}{2}\right)}{1 - \tan^2\left(\frac{x}{2}\right)} = \frac{2t}{1 - t^2} \Rightarrow$$

$$\cot(x) = \frac{1}{\tan(x)} = \frac{1 - t^2}{2t}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha \Rightarrow$$

$$\sin x = 2 \sin \frac{x}{2} \cdot \cos \frac{x}{2}$$

$$\frac{\sin x}{1} = 2 \sin \frac{x}{2} \cdot \cos \frac{x}{2}$$

$$\begin{aligned} & \tan \frac{x}{2} = \frac{\sin \frac{x}{2} \cdot \cos \frac{x}{2}}{\cos^2 \frac{x}{2}} = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \Rightarrow \\ & \tan^2 \frac{x}{2} = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \Rightarrow \\ & \tan^2 \frac{x}{2} = 2t \end{aligned}$$

$$\sin x = \frac{2t}{1+t^2}$$

$$\cos(2x) = \cos^2 x - \sin^2 x \Rightarrow$$

$$\cos x = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}$$

$$\begin{aligned} \frac{\cos x}{1} &= \frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2}} = \\ &= \frac{\cancel{\cos^2 \frac{x}{2}} - \cancel{\sin^2 \frac{x}{2}}}{\cancel{\cos^2 \frac{x}{2}} + \cancel{\sin^2 \frac{x}{2}}} = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \Rightarrow \\ &= \frac{\cancel{\cos^2 \frac{x}{2}} - \cancel{\sin^2 \frac{x}{2}}}{\cancel{\cos^2 \frac{x}{2}} + \cancel{\sin^2 \frac{x}{2}}} = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \Rightarrow \end{aligned}$$

$$\cos x = \frac{1 - t^2}{1 + t^2}$$

$$t = \tan\left(\frac{x}{2}\right)$$

$$\tan(x) = \frac{1}{\cot(x)} = \frac{1 - t^2}{2t}$$

$$\sin(x) = \frac{2t}{1+t^2}$$

$$\cos(x) = \frac{1 - t^2}{1 + t^2}$$

Ф-лы преобр. произвед в сумму:

$$\begin{cases} \cos(x+y) = \cos x \cos y - \sin y \sin x \\ \cos(x-y) = \cos x \cos y + \sin y \sin x \end{cases}$$

$$+ | 2 \cos x \cos y = \cos(x+y) + \cos(x-y)$$

$$\cos x \cos y = \frac{\cos(x+y) + \cos(x-y)}{2}$$

$$- | \cos(x-y) - \cos(x+y) = 2 \sin x \sin y$$

$$\sin x \sin y = \frac{\cos(x-y) - \cos(x+y)}{2}$$

$$\begin{cases} \sin(x-y) = \sin y \cos x - \sin x \cos y \\ \sin(x+y) = \sin y \cos x + \sin x \cos y \end{cases}$$

$$\sin(x-y) + \sin(x+y) = 2 \sin y \cos x$$

$$\sin y \cos x = \frac{\sin(x-y) + \sin(x+y)}{2}$$

Формулы преобр. суммы в произвед.

$$x+y = \alpha \quad x-y = \beta$$

$$y = \alpha - x \quad x = \beta + y$$

$$y = \alpha - \beta - y \Rightarrow 2y = \alpha - \beta$$

$$y = \frac{\alpha - \beta}{2}$$

$$\beta = x - y = x - \frac{\alpha - \beta}{2}$$

$$x = \beta + \frac{\alpha - \beta}{2} = \frac{2\beta + \alpha - \beta}{2} = \frac{\beta + \alpha}{2}$$

$$y = \frac{\alpha - \beta}{2} \quad x = \frac{\alpha + \beta}{2}$$

$$\cos x \cos y = \frac{\cos(x+y) + \cos(x-y)}{2} \Leftrightarrow$$

$$2 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right) = \cos \alpha + \cos \beta$$

$$\cos \alpha + \cos \beta = 2 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)$$

Thema zur Erinnerung:

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cdot \cos \frac{\alpha - \beta}{2}$$

$$\sin \alpha - \sin \beta = 2 \sin \frac{\alpha - \beta}{2} \cos \frac{\alpha + \beta}{2}$$

$$\cos \alpha - \cos \beta = -2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

Формула приведения
тригонометрических

$$a^2 + b^2 > 0$$

$$a \sin x + b \cos x = \sqrt{a^2 + b^2} \left(\underbrace{\frac{a}{\sqrt{a^2 + b^2}}}_{\vec{\alpha}} \sin x + \underbrace{\frac{b}{\sqrt{a^2 + b^2}}}_{\vec{\beta}} \cos x \right) \quad (*)$$

$$\tilde{a}^2 + \tilde{b}^2 = 1 \Rightarrow \exists \varphi : \left(\underbrace{\tilde{a}}_{\cos \varphi}, \underbrace{\tilde{b}}_{\sin \varphi} \right)$$

$$(*) \quad \sqrt{a^2 + b^2} (\cos \varphi \sin x + \sin \varphi \cos x) = \sqrt{a^2 + b^2} \sin(x + \varphi)$$

→ приведение к одному
тригонометрическому
члену.

$$a \sin x + b \cos x = \sqrt{a^2 + b^2} \sin(x + \varphi)$$